

MAT1002 Lecture 1,

Outline

- Infinite Sequences (10.1)
 - ↳ Definition
 - ↳ Limits and convergence
 - ↳ Properties of limits
 - ↳ Connection with functions of real variables
 - ↳ Some common limits
 - ↳ Recursively defined sequences
 - ↳ Bounded sequences

Infinite Sequences

Def: An infinite sequence, or simply sequence, is a list

$$\{a_1, a_2, a_3, \dots, a_n, \dots\}.$$

Remarks

- A sequence can be viewed as a function with domain being the set of positive integers $\{1, 2, 3, \dots\}$, i.e., $a_1 = f(1)$, $a_2 = f(2), \dots$
- The domain $\{1, 2, 3, \dots\}$ is also called the index set of the sequence $\{a_1, a_2, a_3, \dots\}$.
- The index set sometimes can start with a number other than 1, e.g., $\{a_0, a_1, a_2, a_3, \dots\}$ and $\{b_3, b_4, b_5, \dots\}$ are also considered sequences.
- We use $\{a_n\}_{n=1}^{\infty}$ to denote $\{a_1, a_2, a_3, \dots\}$; we may also simply write $\{a_n\}$ if the index set is not important or is clear from the context.

e.g.

$$\{(-1)^{n+1}n^2\}_{n=1}^{\infty} = \{1, -4, 9, -16, 25, -36, \dots\}$$

• A sequence can be defined recursively: the rules

$$F_0 := 0, F_1 := 1, F_i := F_{i-1} + F_{i-2} \text{ (for } i \geq 2 \text{)}$$

define the sequence $\{F_n\}_{n=0}^{\infty} = \{0, 1, 1, 2, 3, 5, 8, 13, \dots\}$

(called the **Fibonacci sequence**.)

Limits and Convergence

Definition

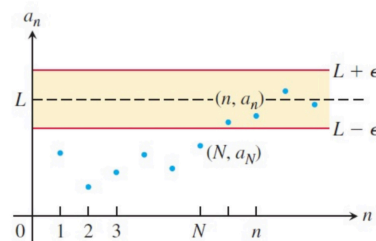
A sequence $\{a_n\}$ is said to be **convergent** if there exists $L \in \mathbb{R}$ satisfying the following property: for all $\epsilon > 0$, there exists N such that $|a_n - L| < \epsilon$ for all $n > N$. In this case, we write

$$\lim_{n \rightarrow \infty} a_n = L, \quad \text{or} \quad a_n \rightarrow L \quad (\text{as } n \rightarrow \infty),$$

and say that $\{a_n\}$ **converges** to L . The number L is called the **limit** of $\{a_n\}$. A sequence that is not convergent is said to be **divergent**.

Remark

The choice of N depends on ϵ , in general.
(Bigger N may be required for smaller ϵ .)



e.g. Show that $\{\frac{1}{n}\}_{n=1}^{\infty}$ converges using definition.

Proof: Let $\epsilon > 0$ be arbitrary. If we pick N to satisfy

$N > \frac{1}{\varepsilon}$, then for all $n > N$, we have

$$|a_n - 0| = \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \frac{1}{N} < \varepsilon,$$

so $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, i.e., $\{\frac{1}{n}\}$ converges to 0. □

e.g. Show that $\{a_n\}_{n=1}^{\infty}$ converges using definition, where

$$a_n := \frac{n}{2n+1}.$$

Solution: To be shown in class.

e.g. Show that $\{(-1)^n\}_{n=0}^{\infty}$ diverges.

Proof: Suppose not. Then $\lim_{n \rightarrow \infty} (-1)^n = L$ for some $L \in \mathbb{R}$.

Let $\varepsilon = 1$. By definition, there exists N such that for all $n > N$, we have $|(-1)^n - L| < 1$. In particular, when

n is even, we have $|L - 1| < 1$, which implies

$$0 < L < 2.$$

On the other hand, when n is odd, we have $|L - (-1)| < 1$, which implies

$$-2 < L < 0.$$

Such an L does not exist, contradiction. Hence $\{(-1)^n\}$ diverges.



Remark

Adding finitely many terms to a sequence does not change its convergence/divergence, e.g.

- $\{7, 48, 56, -\pi, 4, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \dots\}$ Converges
- $\{46, 3, e^{-1}, 12, -1, 1, -1, 1, -1, 1, \dots\}$ diverges.

Definition

Let $\{a_n\}$ be a sequence of real numbers.

- ▶ We write $\lim_{n \rightarrow \infty} a_n = \infty$, or simply $a_n \rightarrow \infty$ (as $n \rightarrow \infty$), if for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that $a_n > M$ for all n satisfying ~~$n \geq N$~~ $n > N$.
- ▶ We write $\lim_{n \rightarrow \infty} a_n = -\infty$, or simply $a_n \rightarrow -\infty$ (as $n \rightarrow \infty$), if for every $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that $a_n < M$ for all n satisfying ~~$n \geq N$~~ $n > N$.

If $\lim_{n \rightarrow \infty} a_n = \infty$, then $\{a_n\}$ is said to diverge to ∞ .

If $\lim_{n \rightarrow \infty} a_n = -\infty$, then $\{a_n\}$ is said to diverge to $-\infty$.

e.g.

- $\{\sqrt{n}\}$ diverges to ∞ .
- $\{(-1)^n n\}$ diverges, but not to $\pm\infty$.

Properties of Sequence Limits

The following are some properties of sequence limits similar to those of function limits.

THEOREM 10.1.1 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers, and let A and B be real numbers. The following rules hold if $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$.

- | | |
|----------------------------|---|
| 1. Sum Rule: | $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$ |
| 2. Difference Rule: | $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$ |
| 3. Constant Multiple Rule: | $\lim_{n \rightarrow \infty} (k \cdot b_n) = k \cdot B$ (any number k) |
| 4. Product Rule: | $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = A \cdot B$ |
| 5. Quotient Rule: | $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$ |

$$\begin{aligned} \text{e.g. } \lim_{n \rightarrow \infty} \frac{n^2 + 1}{n^3 + 3} &= \lim_{n \rightarrow \infty} \frac{\frac{1}{n} + \frac{1}{n^3}}{1 + \frac{3}{n^3}} = \frac{\lim_{n \rightarrow \infty} \frac{1}{n} + \lim_{n \rightarrow \infty} \frac{1}{n^3}}{\lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} \frac{3}{n^3}} \\ &= \frac{\lim_{n \rightarrow \infty} \frac{1}{n} + \left(\lim_{n \rightarrow \infty} \frac{1}{n}\right)^3}{\lim_{n \rightarrow \infty} 1 + 3\left(\lim_{n \rightarrow \infty} \frac{1}{n}\right)^3} = \frac{0 + 0^3}{1 + 3 \cdot 0^3} = 0. \end{aligned}$$

e.g. If $\{a_n\}$ diverges, then so does $\{ca_n\}$ for all constant $c \neq 0$. Indeed, if $ca_n \rightarrow L$ as $n \rightarrow \infty$, then $a_n = \frac{1}{c}(ca_n) \rightarrow \frac{1}{c}L$, implying that $\{a_n\}$ converges.

The following theorem connects sequences with continuous function with real variables.

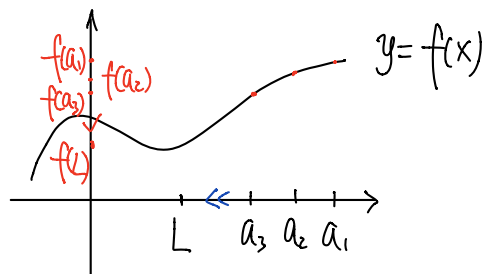
10.1.3

THEOREM 10.1.3 — The Continuous Function Theorem for Sequences Let $\{a_n\}$ be a sequence of real numbers. If $a_n \rightarrow L$ and if f is a function that is continuous at L and defined at all a_n , then $f(a_n) \rightarrow f(L)$.

Note that $\{f(a_n)\}$ is a sequence, and the theorem is saying that the sequence limit of $\{f(a_n)\}$ is $f(L)$.

Proof: Let $\varepsilon > 0$ be arbitrary.

• Since f is continuous at L , there exists $\delta > 0$ such that



if $|x - L| < \delta$ then $|f(x) - f(L)| < \varepsilon$. ①

" x close enough to L "

" $f(x)$ close enough to $f(L)$ "

• Since $a_n \rightarrow L$, there exists N such that for all $n > N$,

$|a_n - L| < \delta$. ← This is your "new ε "
② for sequence limits.

• Combining ① and ②, we have

$$\forall n > N, |f(a_n) - f(L)| < \varepsilon.$$

• By definition, $f(a_n) \rightarrow f(L)$ as $n \rightarrow \infty$. □

e.g. Find $\lim_{n \rightarrow \infty} 2^{\frac{1}{n}}$.

Sol: The function $f(x) = 2^x$ is continuous. Since $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, $2^{\frac{1}{n}} = f(\frac{1}{n}) \rightarrow f(0) = 2^0 = 1$ as $n \rightarrow \infty$.
 $\therefore \lim_{n \rightarrow \infty} 2^{\frac{1}{n}} = 1$.

10.1.2

THEOREM 2—The Sandwich Theorem for Sequences Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences of real numbers. If $a_n \leq b_n \leq c_n$ holds for all n beyond some index N , and if $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$ also.

The proof of this is similar to the function limit version.
See Chapter 2 of Thomas'.

e.g. Since

$$0 < \frac{n!}{n^n} = \frac{n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1}{n \cdot n \cdot \dots \cdot n \cdot n} \leq \frac{1}{n}, \quad \forall n \geq 1,$$

and $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$.

e.g. If $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$, since

$$-|a_n| \leq a_n \leq |a_n|, \quad \forall n,$$

and $\lim_{n \rightarrow \infty} (|a_n|) = -0 = 0$.

Connection with Functions of Real Variables

10.1.4

THEOREM Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{x \rightarrow \infty} f(x) = L \quad \Rightarrow \quad \lim_{n \rightarrow \infty} a_n = L.$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} f(n) = \lim_{x \rightarrow \infty} f(x)$$

Seq. limit function limit.

Proof: Suppose that $\lim_{x \rightarrow \infty} f(x) = L$. Let $\varepsilon > 0$. Then

$$\exists N \forall x \in \mathbb{R}, \text{ if } x > N \text{ then } |f(x) - L| < \varepsilon.$$

In particular, for all integers n with $n > N$, we have

$$|a_n - L| = |f(n) - L| < \varepsilon.$$

$$\text{Hence } \lim_{n \rightarrow \infty} a_n = L.$$



Theorem 10.1.4 allows one to use techniques for function limits to find sequence limits.

e.g. Find $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$.

$$\begin{aligned} \text{Sol. } \lim_{n \rightarrow \infty} \frac{\ln n}{n} &= \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} \quad (\text{L'Hôpital}) \\ &= 0. \end{aligned}$$

Some Common Limits

10.1.5

THEOREM 5 The following six sequences converge to the limits listed below:

1. $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

2. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

3. $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$

4. $\lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1)$

5. $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad (\text{any } x)$

6. $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad (\text{any } x)$

In Formulas (3) through (6), x remains fixed as $n \rightarrow \infty$.

Proof: 1. ✓

2. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = \lim_{n \rightarrow \infty} n^{\frac{1}{n}} = \lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \lim_{x \rightarrow \infty} e^{\frac{1}{x} \ln x}$

$= e^{\lim_{x \rightarrow \infty} \frac{\ln x}{x}} = e^0 = 1.$

6. Since $\frac{-|x|^n}{n!} \leq \frac{x^n}{n!} \leq \frac{|x|^n}{n!}$ for all x and n , ↖ $\rightarrow 0$

it suffices to show that $\lim_{n \rightarrow \infty} \frac{|x|^n}{n!} = 0$.

Choose N s.t. $N \geq |x|$. Then for all $n > N$,

$$0 \leq \frac{|x|^n}{n!} = \frac{|x|^N}{N!} \underbrace{\frac{|x|}{N+1} \frac{|x|}{N+2} \cdots \frac{|x|}{n}}_{\substack{\leq |x|^N \\ < 1}} \leq |x|^N \cdot \frac{|x|}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence $\lim_{n \rightarrow \infty} \frac{|x|^n}{n!} = 0$ by Sandwich Theorem.

3, 4, 5: see Thomas'.



Recursively Defined Sequences

e.g. (10.1, ex 97) Given that the sequence

$$\left\{ 2, 2 + \frac{1}{2}, 2 + \frac{1}{2 + \frac{1}{2}}, 2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}, \dots \right\}$$

converges, find its limit.

Side: the limit is a "continued fraction", commonly appeared in number theory.)

Sol. • The sequence is given by $a_1 = 2$, $a_{n+1} = 2 + \frac{1}{a_n}$ for $n \geq 1$.

• Let $L := \lim_{n \rightarrow \infty} a_n$. Then $L = \lim_{n \rightarrow \infty} a_{n+1}$.

$$\cdot \quad a_{n+1} = 2 + \frac{1}{a_n} \Rightarrow \lim_{n \rightarrow \infty} a_{n+1} = 2 + \frac{1}{\lim_{n \rightarrow \infty} a_n} \Rightarrow L = 2 + \frac{1}{L}$$

$$\Rightarrow L^2 = 2L + 1 \Rightarrow L^2 - 2L - 1 = 0$$

$$\Rightarrow L = \frac{2 \pm \sqrt{4 + 4}}{2} = 1 \pm \sqrt{2}.$$

• Since $a_n > 0$ for all n , $L \geq 0$. So $L = 1 + \sqrt{2}$.

Bounded Sequences

DEFINITIONS A sequence $\{a_n\}$ is **bounded from above** if there exists a number M such that $a_n \leq M$ for all n . The number M is an **upper bound** for $\{a_n\}$. If M is an upper bound for $\{a_n\}$ but no number less than M is an upper bound for $\{a_n\}$, then M is the **least upper bound** for $\{a_n\}$.

A sequence $\{a_n\}$ is **bounded from below** if there exists a number m such that $a_n \geq m$ for all n . The number m is a **lower bound** for $\{a_n\}$. If m is a lower bound for $\{a_n\}$ but no number greater than m is a lower bound for $\{a_n\}$, then m is the **greatest lower bound** for $\{a_n\}$.

If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is **bounded**. If $\{a_n\}$ is not bounded, then we say that $\{a_n\}$ is an **unbounded** sequence.

Remarks

One can show that $\{a_n\}$ is bounded if and only if $\exists x \in \mathbb{R}$ and $K \in \mathbb{R}$ such that $|a_n - x| \leq K$ for all n .

e.g.

- $\{\frac{1}{n}\}$ is bounded above by 2, bounded below by -48 .

Its least upper bound is 1 and greatest lower bound is 0.

- $\{n\}$ is unbounded.

Theorem If $\{a_n\}$ is convergent, then $\{a_n\}$ is bounded.

Proof: • Let $L := \lim_{n \rightarrow \infty} a_n$.

- Then $\exists N$ s.t. $|a_n - L| < 1$ for all $n > N$.

- Let $K := \max\{|a_1 - L|, |a_2 - L|, \dots, |a_N - L|, 1\}$.

- Then $|a_n - L| \leq K$ for all $n \geq 1$. Hence $\{a_n\}$ is bounded.

